MNTC 313 – Assignment 2 Version A

Tim Cook has hired you to help him in the design of a special edition of the phone called the Q**Phone**. You are asked to estimate the amount of material required, and the cost of materials to complete the phone. The dimensions of the phone's height and width are shown on the diagram below. Assume the phone is internally design with a circuit and parts that costs \$0.0183 per mm^3. Consider a phone's depth of 7.4 mm (**not shown in the diagram**) and corners as quarter circular of radius 15.16 mm. You are asked to estimate the battery cost to complete the phone as well, which uses approximately a 45% of total volume of the phone. The dimensions of the phone's height and width are shown on the diagram below. Assume the phone's battery costs \$ 0.0095 per mm^3.

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Use C/C++ to write a program that calculates the following values, and prints them on the screen:

- **QPhone's surface area**
- **QPhone's volume**
- **QPhone's material cost**
- **QPhone's battery cost**

Your output must look like:

```
iPhone surface area:
28068 mm2
iPhone volume:
91481 mm3
iPhone material cost:
$1674
battery cost:
$391
```

(Include a new line "\n" at the end of the last line)

Note: **Please refer to the Gradescope test cases for the desired output format.** A starter code file has been provided on OnQ. Download the file, and copy and paste its contents into a new .c/.cpp file in your IDE. Keep distance units in millimeters throughout. Print values to the nearest whole number. Use pi = 3.14.

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